

CYBR Z Technical Challenge #01

The Tunnel Is Up. Are We Safe?

Scenario

A remote access session associated with an employee account has been identified for review.

The VPN session was successfully established and access to internal resources followed.

Your job is not to assume that the session is malicious, and not to assume that successful authentication makes it legitimate.

You are the analyst responsible for determining what the available evidence actually supports.

Your Environment

You will receive an individual WireGuard configuration separately.

After connecting, confirm that the CYBR Z internal lab service is reachable:

`http://10.66.0.1:8080`

Your VPN connection and the supplied evidence are part of the challenge.

Do not perform network discovery, scanning or exploitation.

Phase 1 — Investigate

Your initial evidence package contains:

- E01 — VPN outer traffic
- E02 — VPN inner traffic
- E03 — VPN authentication activity
- E04 — MFA activity
- E05 — Internal access activity
- E06 — User baseline information
- E07 — Endpoint context

Analyse the sources together rather than treating any single artefact as definitive.

Build a timeline and determine what you believe happened.

Initial Verdict

Choose one:

- Legitimate
- Suspicious
- Compromised

Then assign a confidence level:

- Low
- Medium
- High

Your conclusion must explain:

1. What evidence supports your verdict?
2. What evidence contradicts or weakens it?
3. What remains unknown?
4. Which events are most significant?
5. What would you investigate next if additional telemetry were available?

Do not score evidence simply by volume. One strong contradiction may matter more than several weak indicators.

Phase 2 — New Evidence

Additional evidence may be released after your initial analysis.

When that happens, do not rewrite your first conclusion as though you knew the new information from the beginning.

Record your original position, review the new evidence and choose one:

- Maintain
- Revise
- Withdraw

Explain exactly what changed — or why your original conclusion still stands.

The ability to change a decision when evidence changes is part of the challenge.

Phase 3 — Architecture Decision

Once you have completed the investigation, move beyond the incident itself.

Decide whether the organisation should rely on:

- VPN
- ZTNA
- Hybrid

Consider at least the following access scenarios:

- Managed remote employee
- BYOD or contractor
- Internal web application
- Legacy non-web service
- Privileged administrator
- Emergency / break-glass access

For each relevant scenario, consider:

- identity assurance
- device trust
- access scope
- least privilege
- session context
- monitoring and detection
- operational limitations
- exceptions that may still require traditional VPN access

There is no value in choosing a technology label without explaining the control model behind it.

Response and Detection

Your recommendation should also address what the organisation should do if similar activity is detected again.

Consider:

- immediate containment
- credential and session response
- MFA controls
- device verification
- access policy
- logging and telemetry
- detection opportunities
- post-incident review

Separate immediate incident response from longer-term architectural improvement.

Final Presentation

Prepare a maximum of four slides.

Slide 1 — What Happened?

Your timeline, verdict and confidence.

Slide 2 — What Does the Evidence Tell Us?

The strongest evidence, contradictions and remaining gaps.

Slide 3 — VPN, ZTNA or Hybrid?

Your architecture decision and reasoning.

Slide 4 — What Should We Do Next?

Immediate response, detection improvements and longer-term actions.

Presentation time:

5 minutes maximum

A short jury / CTO Q&A will follow separately. Your slides should support your reasoning, not replace it.

What We Are Evaluating

We are interested in how you:

- analyse evidence
- distinguish observation from assumption
- handle uncertainty
- connect multiple telemetry sources
- revise a judgement when new evidence appears
- make a defensible security architecture decision
- communicate technical findings clearly

This challenge is not about finding a hidden sentence that gives you the answer.

It is about making a decision when the evidence is incomplete.

Final Reminder

A secure tunnel protects traffic in transit.

That alone does not tell you who is really on the other side, whether the endpoint should be trusted, or what that identity should be allowed to reach.

The tunnel is up.

Are we safe?